

Experiment 5: Vacuum Cleaner Problem

Aim

To implement the Vacuum Cleaner problem using Artificial Intelligence.

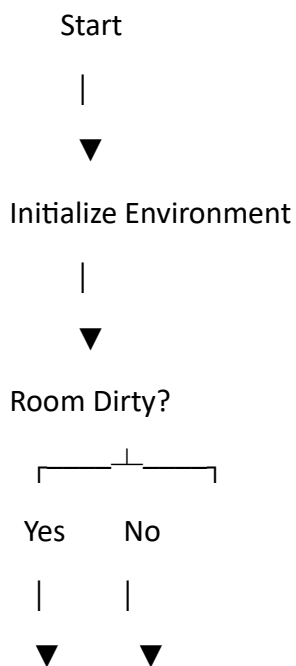
Objective

To design an intelligent vacuum cleaner agent that cleans all dirty rooms with minimum actions.

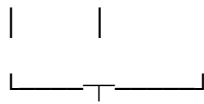
Algorithm

1. Start in the initial room.
2. Check whether the current room is dirty.
3. If dirty, clean the room.
4. Move to the next room.
5. Repeat until all rooms are clean.
6. Display the cleaning sequence

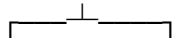
Flowchart



Clean Move Next



All Rooms Clean?



Yes No



Stop Repeat

Python Code

```
rooms = {
```

```
    'A': 'Dirty',
```

```
    'B': 'Dirty'
```

```
}
```

```
for room in rooms:
```

```
    print("Current Room:", room)
```

```
    if rooms[room] == "Dirty":
```

```
        print("Cleaning Room", room)
```

```
        rooms[room] = "Clean"
```

```
    print("Status:", rooms[room])
```

```
    print()
```

```
print("Final State")
```

```
print(rooms)
```

Sample Output

Current Room: A

Cleaning Room A

Status: Clean

Current Room: B

Cleaning Room B

Status: Clean

Final State

{'A': 'Clean', 'B': 'Clean'}

Result

Thus, the Vacuum Cleaner problem was implemented successfully and all rooms were cleaned.